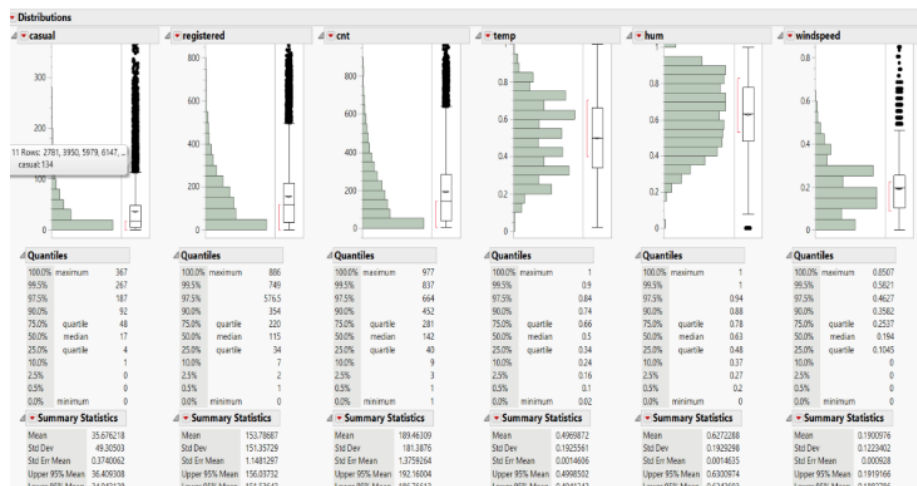


Data Storytelling with JMP

Understanding User Behavior and Environmental Drivers of Bike-Sharing Demand

Two user groups. Different behaviors. One shared environment.



Analytical focus: A visual exploration of how user type, time, seasonality, and weather conditions shape bike-sharing demand.

Tool: JMP **Methods:** Exploratory Data Analysis, Data Visualization, Regression Analysis, and Data Storytelling

1. Who Drives the Demand?

User Type Distribution



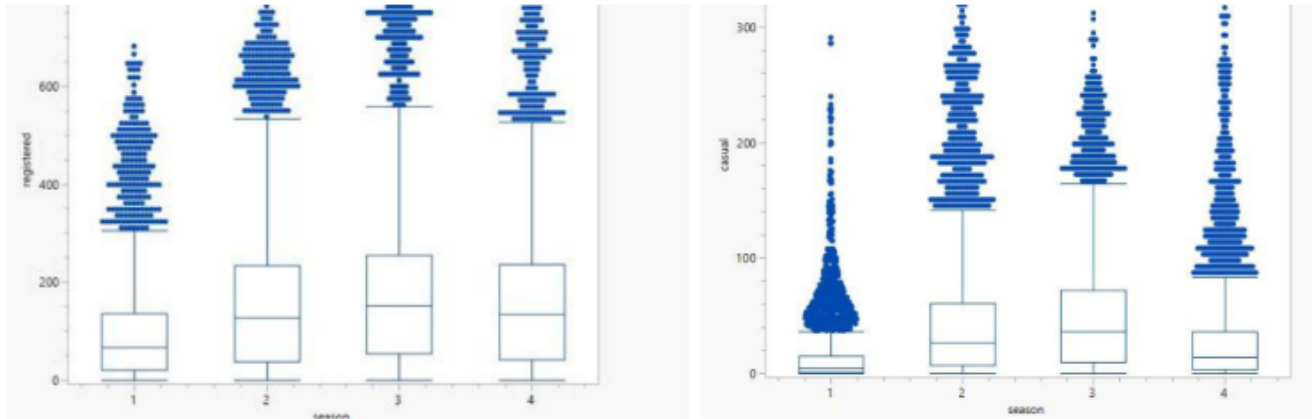
The distribution of demand reveals two distinct user profiles. Casual users show more fluctuating and context-dependent behavior, while Registered users account for a substantially larger and more stable share of system usage.

Key takeaway: Registered users provide the system's demand stability, while Casual users contribute more variable, context-sensitive demand.

2. Seasonality Changes Behavior

Seasonal Demand by User Type

Seasonal Demand by User Type



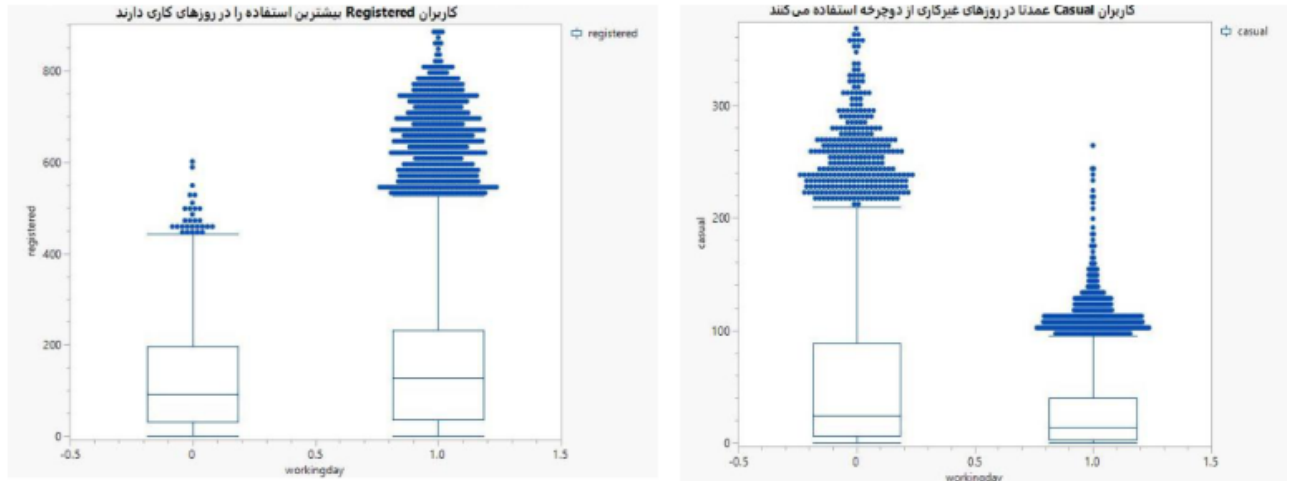
Demand increases during warmer seasons for both user groups. The response is stronger among Casual users, while Registered users remain comparatively active across seasons.

Key takeaway: Seasonality amplifies Casual demand more strongly, while Registered demand remains comparatively resilient.

3. One System, Two Different Purposes

Working Day vs. Non-Working Day Demand

Working Day vs. Non-Working Day Demand



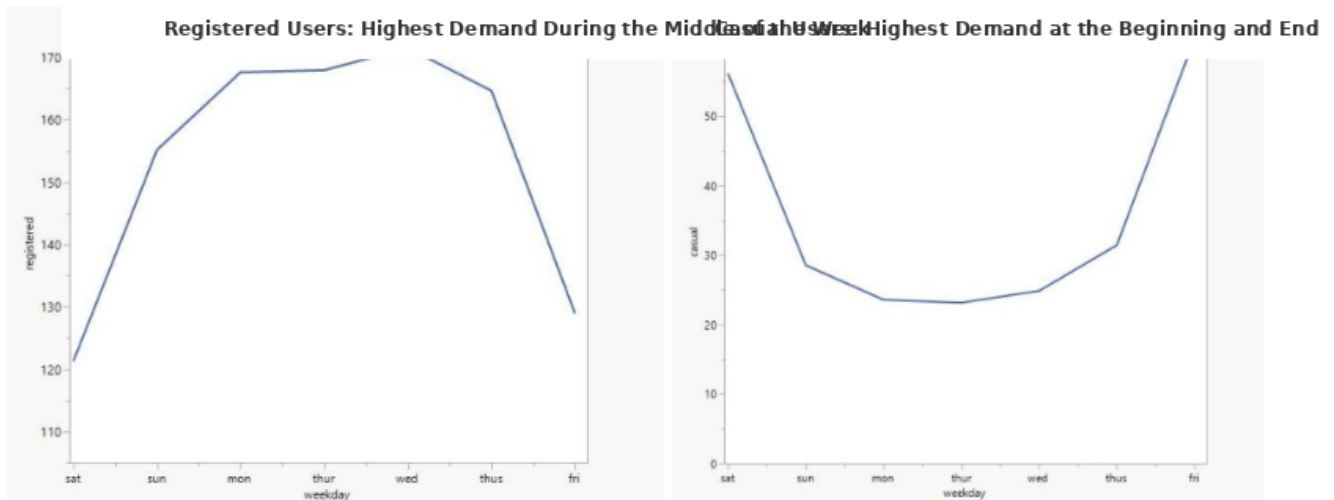
Casual users are more active on non-working days, when leisure time creates opportunities for recreational cycling. Registered users show stronger demand on working days, consistent with routine transportation needs.

Key takeaway: The calendar separates recreational use from routine mobility.

4. Weekly Patterns Tell the Same Story

Weekly Demand Patterns by User Type

Weekly Demand Patterns by User Type



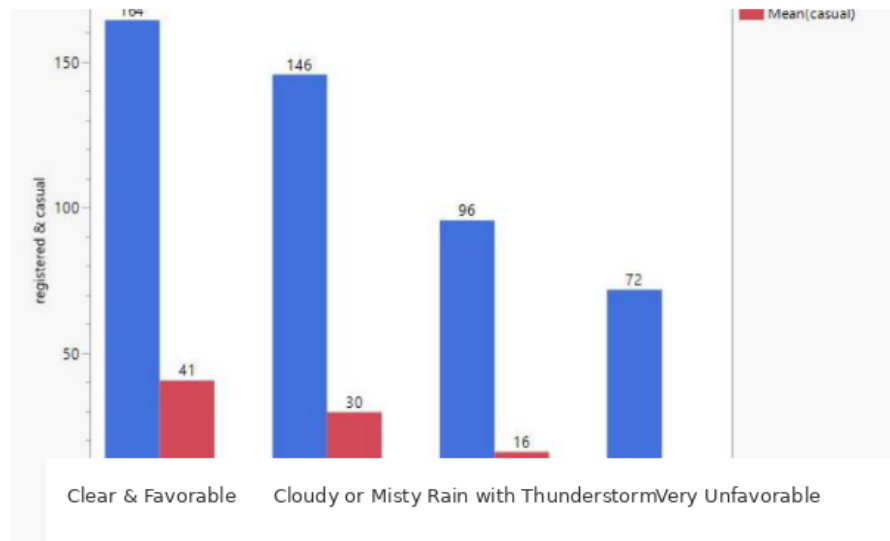
Casual demand is stronger around the beginning and end of the week, while Registered demand reaches higher levels during the middle of the working week.

Key takeaway: Weekly demand patterns reinforce the distinction between leisure-oriented and routine-oriented users.

5. One Weather System, Two Different Reactions

Weather Condition and User Demand

Weather Condition and User Demand



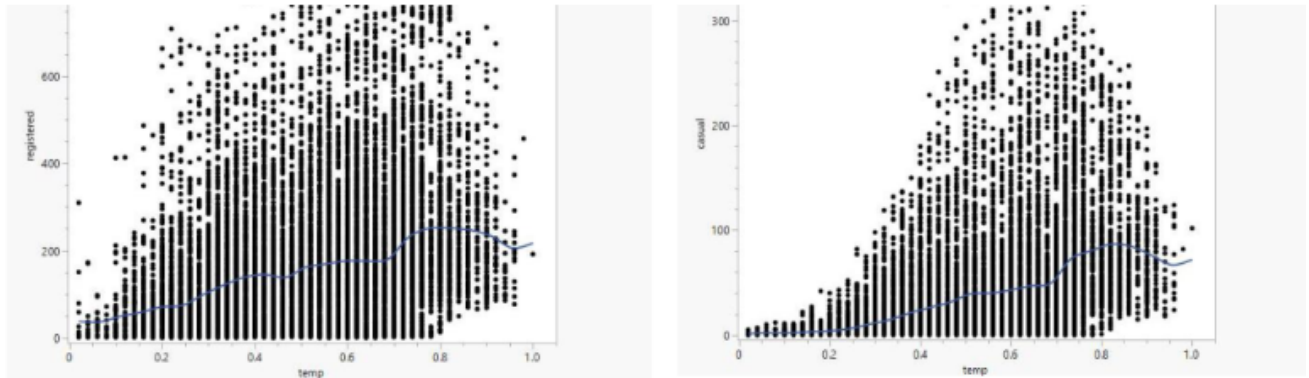
Both user groups prefer favorable weather conditions. However, the decline in Casual usage is more pronounced as conditions become less comfortable. Registered users are more resilient.

Key takeaway: Casual users are more weather-sensitive; Registered users show greater behavioral resilience.

6. Temperature: The Strongest Driver

Temperature and Bike-Sharing Demand

Temperature and Bike-Sharing Demand



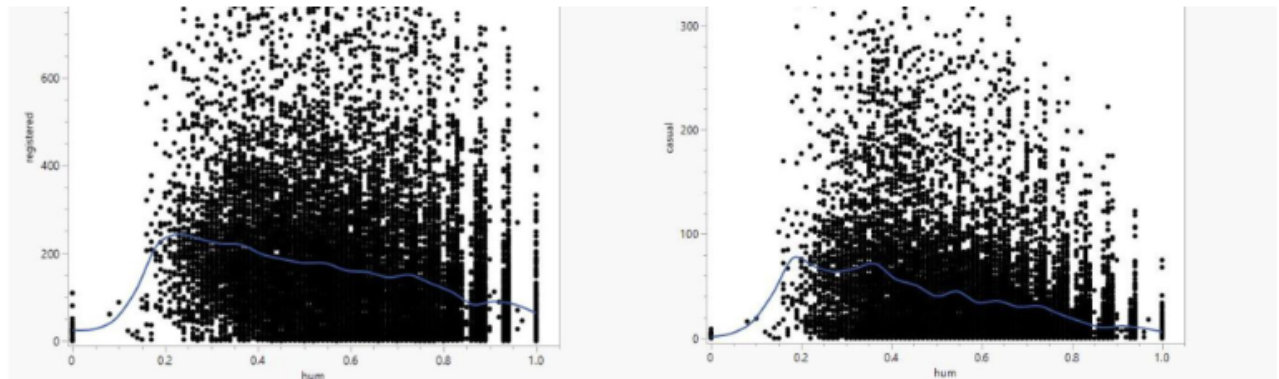
Demand generally increases as temperatures become more favorable for cycling, reaching stronger levels under moderate to warm conditions. At very high temperatures, demand may begin to decline.

Key takeaway: Temperature is a major demand driver, with a stronger behavioral effect on Casual users.

7. Humidity: A Barrier to Comfortable Cycling

Humidity and Bike-Sharing Demand

Humidity and Bike-Sharing Demand



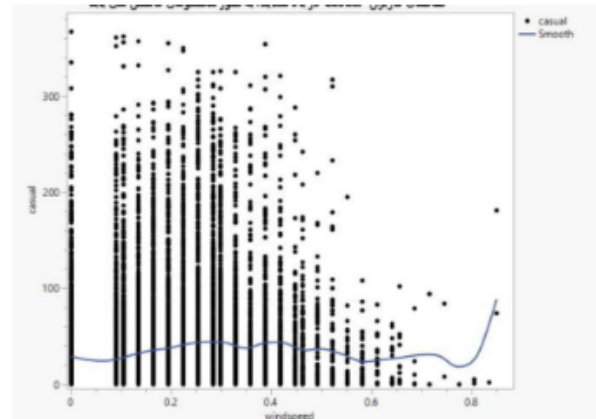
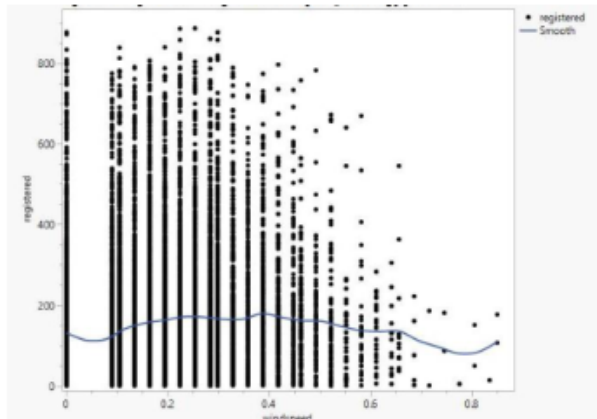
Higher humidity is associated with lower bike-sharing usage, particularly among Casual users. Environmental comfort matters more for discretionary, leisure-oriented trips.

Key takeaway: High humidity discourages usage, especially among users whose trips are more discretionary.

8. Wind Speed: A Weaker but Negative Influence

Wind Speed and Bike-Sharing Demand

Wind Speed and Bike-Sharing Demand



Demand generally decreases as wind speed increases. Casual users respond more strongly to this deterioration in riding conditions, while Registered users show a more gradual decline.

Key takeaway: Wind is a negative influence, but its explanatory power is weaker than temperature and humidity.

9. Comparing Environmental Drivers

Environmental Drivers of Bike-Sharing Demand

Environmental Drivers of Bike-Sharing Demand



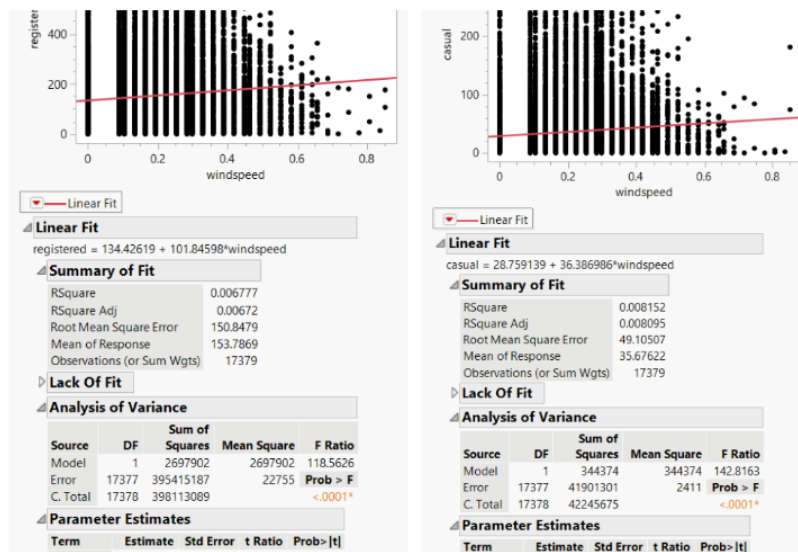
The visual comparison points to a clear hierarchy: temperature shows the strongest relationship with demand, followed by humidity. Wind speed has the weakest explanatory relationship among the three environmental variables examined.

Key takeaway: Temperature leads, humidity follows, and wind speed contributes the least.

10. The Evidence

Regression Analysis: Temperature vs. Demand

Regression Analysis: Temperature vs. Demand

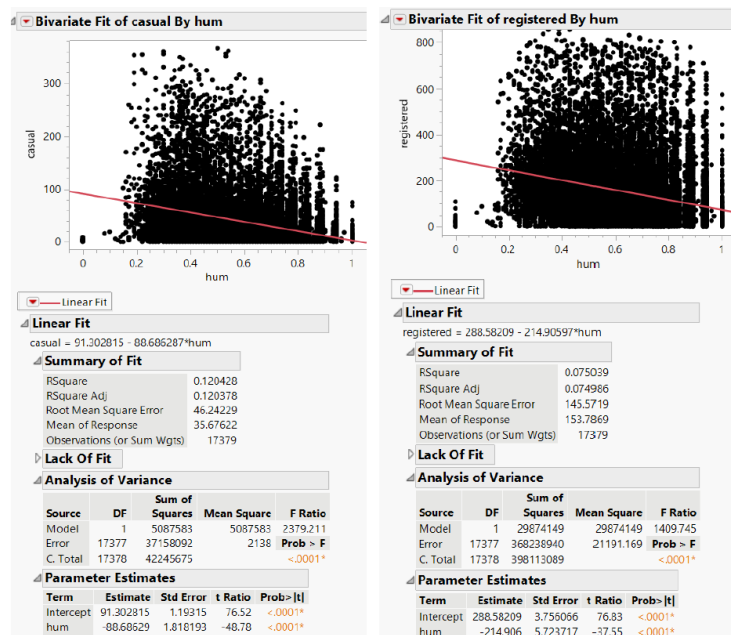


The regression output provides statistical support for the visual pattern. Temperature explains the largest share of variation in Casual user demand among the environmental variables analyzed.

Key takeaway: Temperature is the most influential environmental driver in this analysis.

10. The Evidence

Regression Analysis: Humidity vs. Demand



Humidity also shows a meaningful relationship with demand, although its explanatory power is lower than that of temperature.

Key takeaway: Humidity is an important secondary environmental driver.

10. The Evidence: Comparing Environmental Drivers

Regression Analysis: Environmental Drivers of Bike-Sharing Demand

Environmental Factor	R ²	Approx. Variation Explained
Temperature	0.211	21.1%
Humidity	0.120	12.0%
Wind Speed	0.008	0.8%

Temperature (R² = 0.211): explains approximately 21% of the variation in Casual user demand.

Humidity (R² = 0.120): explains approximately 12% of the variation.

Wind Speed (R² = 0.008): explains less than 1% of the variation.

Interpretation: Temperature is the strongest environmental driver, followed by humidity, while wind speed has the weakest explanatory power.

11. Final Takeaway

The data tells a story of purpose, routine, and environmental sensitivity.

Bike-sharing demand is not driven by a single factor. It reflects the interaction between user purpose, time, seasonality, and environmental conditions.

Registered users: provide greater demand stability and show stronger alignment with routine transportation.

Casual users: are more sensitive to seasonality and environmental comfort.

Temperature: is the strongest environmental driver in the analyzed data.

Humidity: also influences demand, particularly for Casual users.

Wind speed: has the weakest explanatory power among the environmental factors examined.

*The central insight is simple: **the same environment can produce very different behaviors depending on why people use the system.***

Analysis performed with JMP